**Electronic Devices & Circuits Laboratory**

# Experiment 6: NMOS Source Follower with Current-Mirror Load

## 1. Objective

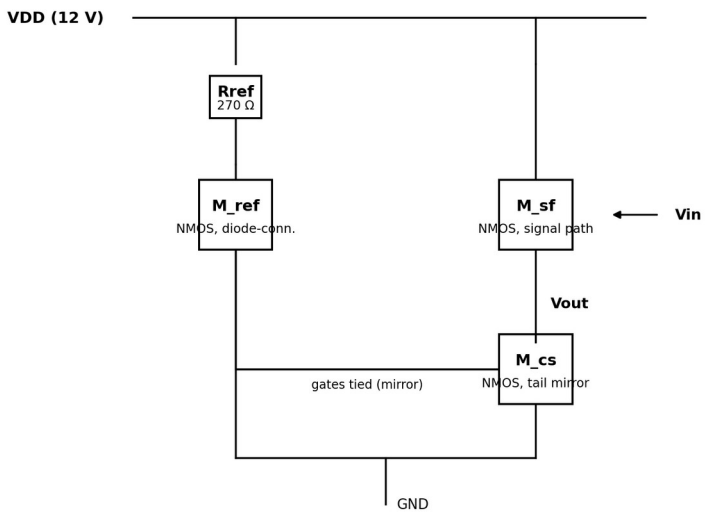
To assemble a simple NMOS source follower (common-drain amplifier) biased by an active current-source load built from a current mirror, and to characterize its DC transfer function, small-signal voltage gain, and output resistance — and to verify that a current-source-loaded follower gives near-unity gain and very low output resistance even at low bias current.

## 2. Components and Equipment Required

| Item | Specification | Quantity |
|---|---|---|
| MOSFET | SI9926BDY, dual N-channel, SO-8, with SO-8-to-DIP breakout adapter (3 of the 4 available channels used: M_ref, M_cs, M_sf) | 2 |
| Reference resistor Rref | 270 Ω, ±5% | 1 |
| Potentiometer | 10 kΩ, multi-turn, for Vin DC bias | 1 |
| DC decoupling resistor R_bias | 1 MΩ (isolates the AC signal path from the bias pot, as in Experiment 5) | 1 |
| Coupling capacitor Cin | 1 µF, AC-couples the function generator onto the DC-biased gate | 1 |
| Load resistors (for Rout measurement) | 10 Ω, 47 Ω, 100 Ω | 1 each |
| DC power supply | 0–30 V regulated, current-limited | 1 |
| Function generator | Sine output, small amplitude, low output impedance, 10 Hz – 100 kHz sweep range | 1 |
| Oscilloscope | 2-channel, to view Vin and Vout | 1 |
| Digital multimeter | For DC transfer measurements | 1–2 |
| Breadboard and jumper wires | — | as required |
| Datasheet | SI9926BDY, Vishay Siliconix (provided copy) | 1 |

## 3. Circuit Diagram

M_ref (diode-connected) and Rref set a reference current Iref ≈ 40 mA. M_ref's gate drives M_cs, mirroring the same current 1:1 to form an active current-source load at the output node. M_sf, the source follower, has its drain tied to VDD and its source tied to this current-source load; the output is taken at the shared M_sf/M_cs node. For the AC measurements (Sections 6.3–6.5), the function generator is coupled onto Vin through a series 1 µF capacitor, with a 1 MΩ R_bias carrying the DC bias from the potentiometer to the gate — the same AC-coupling technique used in Experiment 5, so the DC operating point set in Section 6.2 is undisturbed by the AC source.




## 4. Theory

Because M_cs's gate is fixed (tied to the mirror reference), the current it sinks is fixed at Iref regardless of Vin. Since M_sf and M_cs are the only two current paths into the output node, M_sf must also carry exactly this same fixed current at all times — which means M_sf's own VGS is forced to a single fixed value, independent of Vin. The DC transfer characteristic is therefore simply:

*Vout = Vin − VGS(M_sf), with VGS(M_sf) fixed by Iref — an exact unity-slope, level-shifted line*

This holds across the whole range where M_sf and M_cs both stay in saturation — for this bias point, from about 1.1 V up to the supply rail, a far wider linear range than the differential amplifier's roughly ±150 mV (Experiment 4). The small-signal voltage gain follows the same reasoning: Av = dVout/dVin = gm/(gm + 1/ro,total) ≈ 1 for high gm and high output resistance of the current-source load; any measured gain below 1 comes from the current source's finite output resistance and body effect, neither of which is characterized on the SI9926BDY datasheet, so this gap is left as something to measure rather than predict precisely.

The output resistance looking back into Vout is approximately Rout ≈ 1/gm (ignoring body effect and the current source's own output resistance, both of which raise it slightly). Because this device sits in weak inversion at this bias current (as established in Experiment 4), gm = Iref/(nVT) is unusually large for the current level, so Rout comes out remarkably small — on the order of 1 Ω. This is the classic benefit of a source follower: it presents a low-impedance output regardless of how modest the bias current is.

Frequency response: an exact bandwidth prediction would need the gate and output capacitances under these specific bias and loading conditions, which the datasheet does not characterize (it only gives Ciss/Coss at large power-device currents, not at this weak-inversion bias). Qualitatively, though, two effects point the same way: the very low Rout means the output side is unlikely to show any roll-off within a normal bench sweep,

and the source follower's own gain mechanism does not rely on a high-impedance drain node the way the Experiment 4 amplifier's does. Any roll-off actually observed within the swept range is therefore more likely to come from the input side — the function generator's source resistance and the coupling network interacting with the gate capacitance — than from the follower itself, and is worth identifying rather than assuming away.

## 5. Safety Precautions

- The SI9926BDY has a low gate threshold (VGS(th) = 0.6–1.5 V, min–max per datasheet) and is ESD-sensitive; ground yourself before handling the breakout boards.
- Set the power supply current limit to about 150 mA before connecting the circuit for the first time.
- Do not exceed VGS = ±12 V or VDS = 20 V under any circumstance.
- Verify all pin connections twice against the datasheet before switching on the supply, and confirm which of the two SI9926BDY packages' channels is used for M_ref, M_cs, and M_sf before wiring.
- Always power off before changing any wiring on the breadboard, including when swapping the load resistor for the output-resistance measurement.

## 6. Procedure

### 6.1 Mirror verification

- Assemble the reference branch (Rref, M_ref) and the mirror device M_cs only, with M_sf's gate left open (M_sf not yet connected). Switch on VDD = 12 V with the current limit set to about 150 mA.
- Measure the voltage across Rref and compute Iref = VRref/Rref; confirm Iref ≈ 40 mA.

### 6.2 DC transfer characteristic

- Connect M_sf (drain to VDD, source to the M_cs drain node = Vout). Using the potentiometer, sweep Vin from about 1.5 V to 10.5 V in 1 V steps, recording Vout at each step.
- Verify the slope (ΔVout/ΔVin) is close to 1 and identify the lowest Vin at which the linear relationship breaks down (M_cs leaving saturation).

### 6.3 Small-signal gain

- Set Vin's DC bias to about 7 V (mid-range). Add the Cin coupling capacitor and R_bias as shown in Section 3. Apply a small sine signal (start at ~100 mV amplitude, 1 kHz) from the function generator and measure the amplitude of Vin and Vout on the oscilloscope. Compute Av = Vout(amplitude)/Vin(amplitude).

### 6.4 Frequency response

- With the same setup, sweep the function generator frequency from about 10 Hz to 100 kHz (e.g., 10, 30, 100, 300, 1 k, 3 k, 10 k, 30 k, 100 k Hz), recording the Vout amplitude at each frequency while keeping Vin's amplitude constant.

- Identify any frequency at which the gain drops noticeably from its mid-band value, and note whether it is closer to the low end or high end of the sweep — this helps distinguish an input-coupling effect from anything else.

### 6.5 Output resistance

- At 1 kHz, connect each load resistor (10 Ω, 47 Ω, 100 Ω in turn) from Vout to ground, and record the resulting Vout amplitude for each. Power off between changes.
- From the loaded-versus-unloaded amplitude ratio at each RL, extract Rout using RL/(RL+Rout) = Vout,loaded/Vout,unloaded, and average the three estimates.

## 7. Observation Tables

Table 1 — DC transfer characteristic

| Vin (V) | Vout (V) | Vin − Vout (V) |
|---|---|---|
| 1.5 | … | … |
| 2.5 | … | … |
| 3.5 | … | … |
| 4.5 | … | … |
| 5.5 | … | … |
| 6.5 | … | … |
| 7.5 | … | … |
| 8.5 | … | … |
| 9.5 | … | … |
| 10.5 | … | … |

Table 2 — Frequency response

| Frequency (Hz) | Vout amplitude (mV) | Gain \|Av\| | Gain (dB) |
|---|---|---|---|
| … | … | … | … |

Table 3 — Output resistance measurement

| RL (Ω) | Vout amplitude (mV) | Vout,loaded / Vout,unloaded |
|---|---|---|
| — (unloaded) | … | 1.000 |
| 10 | … | … |
| 47 | … | … |
| 100 | … | … |

## 8. Graphs to Be Plotted

- Vout versus Vin (Table 1) — verify the slope is close to 1 across the linear region and mark where it breaks down.

- Gain (dB) versus frequency (log X axis), from Table 2 — identify any roll-off and its approximate corner frequency.
- Vout,loaded/Vout,unloaded versus RL (Table 3), overlaid with the predicted curve RL/(RL+Rout) using the theoretical Rout from Section 9.

## 9. Theoretical Predictions

Using Iref ≈ 40.4 mA (Rref = 270 Ω, Vth ≈ 1.05 V midpoint), VGS(M_sf) ≈ 1.09 V, gm ≈ 1.04 A/V, Rout ≈ 1/gm ≈ 0.96 Ω:

| Vin (V) | Predicted Vout (V) |
|---|---|
| 1.5 | 0.41 |
| 2.5 | 1.41 |
| 3.5 | 2.41 |
| 4.5 | 3.41 |
| 5.5 | 4.41 |
| 6.5 | 5.41 |
| 7.5 | 6.41 |
| 8.5 | 7.41 |
| 9.5 | 8.41 |
| 10.5 | 9.41 |

Predicted output-resistance loading effect:

| RL (Ω) | Predicted Vout,loaded/Vout,unloaded |
|---|---|
| 10 | 0.912 |
| 47 | 0.980 |
| 100 | 0.990 |

## 10. Calculations — Matching Measured Results with Theory

- From Table 1, compute the measured slope ΔVout/ΔVin over the linear region and compare with the ideal value of 1.
- Compute the measured DC level shift (average Vin − Vout over the linear region) and compare with the predicted VGS(M_sf) ≈ 1.09 V.
- From Section 6.3, compute the measured small-signal Av and compare with the ideal value of 1.
- From Table 2, note the measured mid-band gain and any roll-off frequency observed; state whether it is more consistent with an input-side effect (function generator source resistance and gate capacitance) or something else, per the reasoning in Section 4.
- From Table 3, compute Rout using each RL and average; compare with the predicted ≈ 0.96 Ω.

| Quantity | Predicted | Measured | % deviation |
|---|---|---|---|

| | | | |
|---|---|---|---|
| DC slope ΔVout/ΔVin | 1.00 | … | … |
| Level shift Vin−Vout (V) | 1.09 | … | … |
| Small-signal Av | ≈ 1.00 | … | … |
| Rout (Ω) | 0.96 | … | … |

## 11. Precautions

- Do not exceed the SI9926BDY's VDS (20 V) or VGS (±12 V) ratings at any point.
- Keep the current limit active at about 150 mA throughout.
- Power off before adding or removing the load resistor in Section 6.5.
- Keep the function generator amplitude small enough that Vin's total swing (DC bias ± AC amplitude) stays within the saturation range identified in Section 6.2.